

Topology First-Year Exam, May 2020, Part 2

Work the following problems and show all work. Support all statements to the best of your ability.

1. Show that S^2 is not homeomorphic to S^3 .
2. Prove that the set of irrational numbers $\mathcal{J} \subset \mathbb{R}$ cannot be presented as a countable union of closed sets each of which has empty interior in $\mathcal{J}$.
3. Show that for every continuous map $f : S^1 \rightarrow S^1$ there is $n \in \mathbb{Z}$ such that f is homotopic to $z^n : S^1 \rightarrow S^1$.
4. Show that every map $f : S^2 \rightarrow T$ of the 2-sphere to the torus T is null-homotopic.
5. Show that the torus T is not homeomorphic to the surface of genus 2, $M_2 = T \# T$.

Answer the following with complete definitions or statements or proofs.

- 6. Warning: Suspected error.** The displayed definition is mathematically inconsistent over the stated real domain: $\sqrt{x}$ is not real-valued for negative x . The intended correction to the domain or formula is not uniquely determined, so the source reading has been retained.

Does the sequence $f_n : \mathbb{R} \rightarrow \mathbb{R}$, $f_n(x) = \sqrt{x}/n$, converge in

- (a) the point-wise convergence topology?
- (b) the uniform topology?
- (c) compact open topology?

7. State the Borsuk–Ulam Theorem for S^2 .
8. Is the space I^I separable where $I = [0, 1]$?
9. State the Seifert–van Kampen Theorem.
10. State and prove the Tychonoff Theorem.